

SATYA AKHIL GALLA

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SUMMARY

AI Engineer specializing in End-to-End Generative AI, from data engineering to cloud deployment. Expertise includes architecting scalable ML systems, fine-tuning Large Language Models (Llama-3), and engineering autonomous AI Agents with high-speed reactive loops. Proven researcher with IEEE publications in graph convolutional networks.

EDUCATION

Boston University

Master of Science in Artificial Intelligence

Jan 2026

GPA: 3.7/4.0

IIT Sri City

Bachelor of Technology in Computer Science

May 2024

GPA: 3.5/4.0

TECHNICAL SKILLS

Generative AI & LLMs: Llama-3, Phi-3 (Quantization), RAG Architectures, AI Agents (LangGraph), Neuro-Symbolic AI, Vector DBs

Cloud & Engineering: AWS (EC2/S3), Azure, Docker, CI/CD, FastAPI, SQL, Bash Scripting, Linux, Git, Cursor, GitHub Copilot, Claude Code

Data Science & ML: Pandas, NumPy, Scikit-learn, OpenCV

EXPERIENCE

ISRO (Indian Space Research Organization)

Dec 2022 – May 2023

Machine Learning Engineer

- Boosted hyperspectral classification accuracy on the Chandrayaan-1 dataset by **13.5%** compared to CNN baselines by engineering a **Graph Convolutional Network (GCN)** utilizing Latent Space projections.
- Devised a **Spectral-Spatial Non-Linearity** graph formulation to handle irregular lunar topography, achieving **91% accuracy** on Chandrayaan-2 data and validating results via an **IEEE WHISPERS 2023** publication.

Aaizel International Tech

Feb 2024 – May 2024

Machine Learning Engineer (Internship)

- Architected a multimodal fusion pipeline (IFCNN) merging IR+RGB satellite imagery and constructed the domain's first remote sensing scene graph dataset.
- Fine-tuned a RelTR Transformer for geospatial relationship extraction and deployed the inference engine to Microsoft Azure, enabling real-time semantic visualization.

Terrafic Inc

June 2023 – Nov 2023

Computer Vision Engineer (Internship)

- Integrated the Segment Anything Model (SAM) and OpenCV to automate maritime feature extraction; delivered a production-ready MVP that secured initial client pilots.
- Implemented Super-Resolution (SR) algorithms (4x/6x) to enhance object detection performance on low-resolution defense mapping data.

Boston University

Sep 2025 – Jan 2026

Graduate Teaching Assistant – CS 365 Foundations of Data Science

- Led technical labs on Python optimization and statistical modeling, mentoring 50+ students in data science fundamentals, including Pandas, NumPy, and algorithmic best practices.
- Facilitated weekly code reviews to enforce production-standard coding styles and debugging techniques.

KEY PROJECTS

Serverless Clinical AWQ Llama Engine | [Llama-3.1](#), [Modal](#), [vLLM](#), [Pinecone](#), [Hugging Face](#) | [GitHub](#) | [HuggingFace Spaces](#)

- Initially prototyped a LangGraph-based agentic pipeline on AWS (EC2/S3) using Docker and Linux environments, later migrating to a stateless, W4A16 AWQ-quantized backend on Modal serverless GPUs to optimize compute availability.
- Architected a 384-dimensional Hybrid RAG pipeline utilizing Dense (BGE-Small) and Sparse (BM25) retrieval; integrated deterministic query expansion and a MiniLM Cross-Encoder to ensure high-precision context retrieval across 10,000 documents in Pinecone.
- LoRA fine-tuned Llama-3.1-8B-Instruct on 34k medical samples achieving final metrics of 0.529s Time-To-First-Token, sub-2s end-to-end latency, 60s cold start time and ~60 tokens/sec throughput on an A10G GPU.

CoCo: Neuro-Symbolic Desktop Companion | [Phi-3](#), [PyTorch](#), [TCP/IPC](#) | [GitHub](#)

- Developed a bio-mimicking Agentic AI architecture to replace static desktop pets, designing a dual-layer system that separates high-speed spinal reflexes (60Hz reactive loop) from slow cortical reasoning (LLM).
- Integrated Phi-3 Mini (4-bit) as the cognitive brain, using asynchronous TCP sockets to fetch complex reasoning without blocking the real-time animation loop.
- Implemented a Leaky Integrate-and-Fire (LIF) neural network inspired model to interpret internal emotional states, enabling the agent to dynamically react to user activity patterns.

Generative Audio: Transformer vs. Diffusion | [AudioLDM](#) | [GitHub](#)

- Conducted a comparative analysis of 8 hybrid architectures, identifying the optimal trade-off between semantic alignment (Transformers) and audio fidelity (Diffusion).
- Engineered a cascaded pipeline achieving a peak CLAP score of 0.35 and FAD of 6.55, validating the superiority of separate structure-texture generation phases.

Multimodal Skin Cancer Detection System | [PyTorch](#) | [GitHub](#)

- Designed a Voting Ensemble using Scikit-learn and XGBoost to fuse CNNs (image) and tabular models (metadata), detecting malignancy in a dataset with extreme 0.1% class imbalance.
- Surpassed single-modal baselines by 15 points (0.96 ROC-AUC) by applying synthetic oversampling (SMOTE) and rigorous cross-validation strategies.

SELECT PUBLICATIONS

Efficient Graph Formulation and Latent Space Integration for Lunar Hyperspectral Image Classification – [IEEE WHISPERS 2023](#) | [Google Scholar](#)

- Proposed a Latent Graph Convolutional Network (GCN) utilizing an Autoencoder for dimensionality reduction, achieving 90.1% accuracy on Chandrayaan-1 benchmarks by engineering a spectral-spatial non-linearity function.

Enhancing Hyperspectral Classification through GCNs with Adaptive Graph Construction – [NASA NESF 2023 \(Poster\)](#)

- Presented a comparative analysis of adaptive graph constructions for Chandrayaan-2 IIRS data, identifying non-linear spectral-spatial connections as optimal for handling irregular lunar topography.